

Course 5073A

5 Credits: 4

Program Elective

Source-grounded

# Petroleum Refining &

□ To impart the basic knowledge on fossil fuels, up-stream and downstream processes

## OFFICIAL SOURCE DECLARATION

### How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 5073A is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

**Source file:** 5073A.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

## COURSE DASHBOARD

### Official course information

#### Course information

| Item             | Official information                        |
|------------------|---------------------------------------------|
| Programme        | Diploma in Chemical Engineering             |
| Course code      | 5073A                                       |
| Course title     | Petroleum Refining &                        |
| Semester         | 5 Credits: 4                                |
| Course category  | Program Elective                            |
| Periods per week | 4 (L: 4 T: 0 P: 0) Periods per semester: 60 |

| Item                 | Official information                                 |
|----------------------|------------------------------------------------------|
| Periods per semester | 60                                                   |
| Credits              | 4                                                    |
| Prerequisite         | See the official syllabus PDF                        |
| Assessment           | Use the official assessment section reproduced below |

## Modules

4 official module sections detected.

## Topic entries

17 source-aligned topic entries retained.

## Study mode

Theory and application emphasis.

## STUDY METHOD

## How to use this lesson

### First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

### Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

### Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

## Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

### LEARNING OUTCOMES

## Objectives, outcomes and mapping

### Course objectives

- □ To impart the basic knowledge on fossil fuels, up-stream and downstream processes for crude petroleum.
- □ To comprehend the manufacturing processes and applications of various petrochemical products.

### Course outcomes

| CO  | Official outcome                                                                             | Study level           |
|-----|----------------------------------------------------------------------------------------------|-----------------------|
| CO1 | 14 Understanding processes of crude petroleum. Illustrate the downstream processes for crude | See official syllabus |
| CO2 | 17 Understanding petroleum. Summarize the treatment techniques for petroleum                 | See official syllabus |
| CO3 | 15 Understanding fractions. Summarize the manufacturing processes and                        | See official syllabus |
| CO4 | 14 Understanding applications of various petrochemicals.                                     | See official syllabus |

### CO-PO mapping evidence

| Row | Extracted official mapping |
|-----|----------------------------|
| CO1 | CO1 2                      |
| CO2 | CO2 2                      |
| CO3 | CO3 2                      |
| CO4 | CO4 3 3                    |

## Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

### Module coverage

| Module   | Official heading                                                       | Topic entries | Hours        |
|----------|------------------------------------------------------------------------|---------------|--------------|
| Module 1 | Summarise fossil fuels and the up-stream processes of crude petroleum. | 4             | As specified |
| Module 2 | Illustrate the downstream processes for crude petroleum.               | 5             | As specified |
| Module 3 | Summarize the treatment techniques for petroleum fractions.            | 4             | As specified |
| Module 4 | petrochemicals.                                                        | 4             | As specified |

### MODULE 1

## Summarise fossil fuels and the up-stream processes of crude petroleum.

This module is presented from the official syllabus scope for Petroleum Refining &. Use the topic cards for learning and the source transcript for exact coverage.

**As specified**  
official hours

**CO-linked**  
see outcomes

### Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

### Key facts

- Official module title: Summarise fossil fuels and the up-stream processes of crude petroleum.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

## Classify fossil fuels. 2 Remembering Summarize the production methods,

Classify fossil fuels. 2 Remembering Summarize the production methods, is an official topic in Module 1 of Petroleum Refining &. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Classify fossil fuels. 2 Remembering Summarize the production methods, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, classify fossil fuels. 2 remembering summarize the production methods, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                                                  |
|---------------------|------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What Classify fossil fuels. 2 Remembering Summarize the production methods, means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                          |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

**Malayalam support:** Module 1- Classify fossil fuels. 2 Remembering Summarize the production methods, definition/meaning .  
 , steps, application . Technical terms English- .

## 4 Understanding characteristics and properties of gaseous fuels. Summarise the characterization and properties of

4 Understanding characteristics and properties of gaseous fuels. Summarise the characterization and properties of is an official topic in Module 1 of Petroleum Refining &. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 4 Understanding characteristics and properties of gaseous fuels. Summarise the characterization and properties of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 4 understanding characteristics and properties of gaseous fuels. summarise the characterization and properties of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                                                                                             |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What 4 Understanding characteristics and properties of gaseous fuels. Summarise the characterization and properties of means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                                                     |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                                                           |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.



**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1-4 Understanding characteristics and properties of gaseous fuels. Summarise the characterization and properties of  $\text{C}_2\text{H}_2$   $\text{C}_2\text{H}_4$  definition/meaning  $\text{C}_2\text{H}_2$   $\text{C}_2\text{H}_4$   $\text{C}_2\text{H}_6$   $\text{C}_2\text{H}_2$   $\text{C}_2\text{H}_4$   $\text{C}_2\text{H}_6$ , steps, application  $\text{C}_2\text{H}_2$   $\text{C}_2\text{H}_4$   $\text{C}_2\text{H}_6$   $\text{C}_2\text{H}_2$   $\text{C}_2\text{H}_4$   $\text{C}_2\text{H}_6$ . Technical terms English- $\text{C}_2\text{H}_2$   $\text{C}_2\text{H}_4$   $\text{C}_2\text{H}_6$ .

## 4 Understanding crude petroleum. Illustrate the upstream processes for crude

4 Understanding crude petroleum. Illustrate the upstream processes for crude is an official topic in Module 1 of Petroleum Refining &. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 4 Understanding crude petroleum. Illustrate the upstream processes for crude using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 4 understanding crude petroleum. illustrate the upstream processes for crude should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                                                        |
|---------------------|------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What 4 Understanding crude petroleum. Illustrate the upstream processes for crude means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                      |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1-4 Understanding crude petroleum. Illustrate the upstream processes for crude **പ്രാഥമിക പരിശുദ്ധീകരണം** definition/meaning **പ്രാഥമിക പരിശുദ്ധീകരണം**. **പ്രാഥമിക പരിശുദ്ധീകരണം**, steps, application **പ്രാഥമിക പരിശുദ്ധീകരണം** **പ്രാഥമിക പരിശുദ്ധീകരണം**. Technical terms English-**പ്രാഥമിക പരിശുദ്ധീകരണം**.

**4 Understanding petroleum. Classification of fuel- Categorize primary & secondary solid, liquid and gaseous fuels with examples. Identify the composition, properties and uses of Natural gas, - Illustrate the production of natural gas - Comprehend the production of shale gas. Origin& formation of petroleum - Classification of crude oil - Properties of crude oil -API gravity and density, viscosity, sulphur content and vapour pressure - Evaluation of crude oil. Exploration and drilling of crude oil - Gas-oil separation - Dehydration - Desalting- Crude stabilization and Enhanced oil recovery - Sources of indigenous crude - Sources of imported crude - Petroleum industries in India**

4 Understanding petroleum. Classification of fuel- Categorize primary & secondary solid, liquid and gaseous fuels with examples. Identify the composition, properties and uses of Natural gas, - Illustrate the production of natural gas - Comprehend the production of shale gas. Origin& formation of petroleum - Classification of crude oil - Properties of crude oil -API gravity and density, viscosity, sulphur content and vapour pressure - Evaluation of crude oil. Exploration and drilling of crude oil - Gas-oil separation - Dehydration - Desalting- Crude stabilization and Enhanced oil recovery - Sources of indigenous crude - Sources of imported crude - Petroleum industries in India is an official topic in Module 1 of Petroleum Refining &. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### **Concept and explanation**

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### **Study points**

- Define or identify 4 Understanding petroleum. Classification of fuel- Categorize primary & secondary solid, liquid and gaseous fuels with examples. Identify the composition, properties and uses of Natural gas, - Illustrate the production of natural gas - Comprehend the production of shale gas. Origin& formation of petroleum - Classification of crude oil - Properties of crude oil -API gravity and density, viscosity, sulphur content and vapour pressure - Evaluation of crude oil. Exploration and drilling of crude oil - Gas-oil separation - Dehydration - Desalting- Crude stabilization and Enhanced oil recovery - Sources of indigenous crude - Sources of imported crude - Petroleum industries in India using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

## Engineering or professional relevance

In Polytechnic practice, 4 understanding petroleum. classification of fuel- categorize primary & secondary solid, liquid and gaseous fuels with examples. identify the composition, properties and uses of natural gas, - illustrate the production of natural gas - comprehend the production of shale gas. origin& formation of petroleum - classification of crude oil - properties of crude oil -api gravity and density, viscosity, sulphur content and vapour pressure - evaluation of crude oil. exploration and drilling of crude oil - gas-oil separation - dehydration - desalting- crude stabilization and enhanced oil recovery - sources of indigenous crude - sources of imported crude - petroleum industries in india should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What 4 Understanding petroleum. Classification of fuel- Categorize primary & secondary solid, liquid and gaseous fuels with examples. Identify the composition, properties and uses of Natural gas, - Illustrate the production of natural gas - Comprehend the production of shale gas. Origin& formation of petroleum - Classification of crude oil - Properties of crude oil -API gravity and density, viscosity, sulphur content and vapour pressure - Evaluation of crude oil. Exploration and drilling of crude oil - Gas-oil separation - Dehydration - Desalting- Crude stabilization and Enhanced oil recovery - Sources of indigenous crude - Sources of imported crude - Petroleum industries in India means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1- 4 Understanding petroleum. Classification of fuel- Categorize primary & secondary solid, liquid and gaseous fuels with examples. Identify the composition, properties and uses of Natural gas, - Illustrate the production of natural gas - Comprehend the production of shale gas. Origin&

formation of petroleum - Classification of crude oil – Properties of crude oil -API gravity and density, viscosity, sulphur content and vapour pressure – Evaluation of crude oil. Exploration and drilling of crude oil - Gas-oil separation – Dehydration - Desalting- Crude stabilization and Enhanced oil recovery – Sources of indigenous crude - Sources of imported crude - Petroleum industries in India   
 definition/meaning . steps, application . Technical terms English-   
 .

## Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

Summarise fossil fuels and the up-stream processes of crude petroleum.

M1.01 Classify fossil fuels. 2

Remembering

Summarize the production methods,

M1.02 4

Understanding

characteristics and properties of gaseous fuels.

Summarise the characterization and properties of

M1.03 4

Understanding

crude petroleum.

Illustrate the upstream processes for crude

M1.04 4

Understanding

petroleum.

Contents:

Classification of fuel- Categorize primary & secondary solid, liquid and gaseous fuels with examples.

Identify the composition, properties and uses of Natural gas, - Illustrate the production of

natural gas - Comprehend the production of shale gas.

Origin& formation of petroleum - Classification of crude oil – Properties of crude oil -API

gravity and density, viscosity, sulphur content and vapour pressure – Evaluation of crude

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam

## MODULE 2

# Illustrate the downstream processes for crude petroleum.

This module is presented from the official syllabus scope for Petroleum Refining &. Use the topic cards for learning and the source transcript for exact coverage.

**As specified**  
official hours

**CO-linked**  
see outcomes

### Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

### Key facts

- Official module title: Illustrate the downstream processes for crude petroleum.
- Official duration: As specified
- Official topic entries captured: 5
- The full source excerpt is preserved below for coverage checking.

## Illustrate modern refinery operations. 1 Understanding Illustrate petroleum products from crude oil by

Illustrate modern refinery operations. 1 Understanding Illustrate petroleum products from crude oil by is an official topic in Module 2 of Petroleum Refining &. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Illustrate modern refinery operations. 1 Understanding Illustrate petroleum products from crude oil by using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, illustrate modern refinery operations. 1 understanding illustrate petroleum products from crude oil by should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                                                                                  |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What Illustrate modern refinery operations. 1 Understanding Illustrate petroleum products from crude oil by means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                                          |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                                                |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.



**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-ഇന്ന് Illustrate modern refinery operations. 1 Understanding Illustrate petroleum products from crude oil by ഇന്റർവ്യൂ ചെയ്ത് definition/meaning ഇന്റർവ്യൂ ചെയ്ത്. ഇന്റർവ്യൂ ചെയ്ത് ഇന്റർവ്യൂ ചെയ്ത്, steps, application ഇന്റർവ്യൂ ചെയ്ത് ഇന്റർവ്യൂ ചെയ്ത് ഇന്റർവ്യൂ ചെയ്ത്. Technical terms English-ഇന്ന് ഇന്റർവ്യൂ ചെയ്ത് ഇന്റർവ്യൂ ചെയ്ത്.

## 4 Understanding fractional distillation. Outline various cracking and reforming processes

4 Understanding fractional distillation. Outline various cracking and reforming processes is an official topic in Module 2 of Petroleum Refining &. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 4 Understanding fractional distillation. Outline various cracking and reforming processes using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 4 understanding fractional distillation. outline various cracking and reforming processes should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                                                                     |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What 4 Understanding fractional distillation. Outline various cracking and reforming processes means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                             |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                                   |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-പാഠ്യ 4 Understanding fractional distillation. Outline various cracking and reforming processes പാഠ്യപദപരിവർത്തനം പാഠ്യപദ definition/meaning പാഠ്യപദപരിവർത്തനം. പാഠ്യപദ പാഠ്യപദ പാഠ്യപദ, steps, application പാഠ്യപദ പാഠ്യപദപരിവർത്തനം. Technical terms English-പാഠ്യപദ പാഠ്യപദപരിവർത്തനം.

## 4 Understanding of petroleum fractions Summarise Visbreaking, Coking, hydro treating

4 Understanding of petroleum fractions Summarise Visbreaking, Coking, hydro treating is an official topic in Module 2 of Petroleum Refining &. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 4 Understanding of petroleum fractions Summarise Visbreaking, Coking, hydro treating using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 4 understanding of petroleum fractions summarise visbreaking, coking, hydro treating should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                                                                |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What 4 Understanding of petroleum fractions Summarise Visbreaking, Coking, hydro treating means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                        |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                              |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-4 Understanding of petroleum fractions  
Summarise Visbreaking, Coking, hydro treating ഏതെങ്കിലും ഒരു വിഭാഗം  
definition/meaning എന്താണ്. ഏതെങ്കിലും ഒരു വിഭാഗം, steps, application  
എന്താണ്. Technical terms English-ൽ എഴുതുക.

## 4 Understanding processes. Summarise Alkylation, Isomerisation and

4 Understanding processes. Summarise Alkylation, Isomerisation and is an official topic in Module 2 of Petroleum Refining &. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 4 Understanding processes. Summarise Alkylation, Isomerisation and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 4 understanding processes. summarise alkylation, isomerisation and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                                              |
|---------------------|--------------------------------------------------------------------------------------------------------------|
| Definition/identity | What 4 Understanding processes. Summarise Alkylation, Isomerisation and means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                      |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                            |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

**Malayalam support:** Module 2-4 Understanding processes. Summarise Alkylolation, Isomerisation and  $\text{C}_{10}\text{H}_8$  definition/meaning  $\text{C}_{10}\text{H}_8$ .  $\text{C}_{10}\text{H}_8$   $\text{C}_{10}\text{H}_8$   $\text{C}_{10}\text{H}_8$ , steps, application  $\text{C}_{10}\text{H}_8$   $\text{C}_{10}\text{H}_8$ . Technical terms English-  $\text{C}_{10}\text{H}_8$   $\text{C}_{10}\text{H}_8$ .

**4 Understanding polymerization processes Block diagram of a modern refinery operation (Including ADU, VDU, gas processing, hydrotreater, FCCU, isomeriser, alkylation, blending, DCU and sulphur recovery unit) Illustrate Crude oil fractionation by distillation- crude preheater- atmospheric distillation unit- vacuum distillation unit-petroleum products - List the petroleum products from crude oil refining and their boiling ranges. Illustrate: Cracking - hydro cracking, thermal cracking, catalytic cracking, fixed bed and fluidized bed - Catalytic reforming Illustrate: Visbreaking operation - Coking - hydro treating Illustrate: Sulphuric acid Alkylation process, Low temperature Isomerisation process and Catalytic polymerization processes.**

4 Understanding polymerization processes Block diagram of a modern refinery operation (Including ADU, VDU, gas processing, hydrotreater, FCCU, isomeriser, alkylation, blending, DCU and sulphur recovery unit) Illustrate Crude oil fractionation by distillation- crude preheater- atmospheric distillation unit- vacuum distillation unit-petroleum products - List the petroleum products from crude oil refining and their boiling ranges. Illustrate: Cracking - hydro cracking, thermal cracking, catalytic cracking, fixed bed and fluidized bed - Catalytic reforming Illustrate: Visbreaking operation - Coking - hydro treating Illustrate: Sulphuric acid Alkylation process, Low temperature Isomerisation process and Catalytic polymerization processes. is an official topic in Module 2 of Petroleum Refining &. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### **Concept and explanation**

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### **Study points**

- Define or identify 4 Understanding polymerization processes Block diagram of a modern refinery operation (Including ADU, VDU, gas processing, hydrotreater, FCCU, isomeriser, alkylation, blending, DCU and sulphur recovery unit) Illustrate Crude oil fractionation by distillation- crude preheater- atmospheric distillation unit- vacuum distillation unit-petroleum products - List the petroleum products from crude oil refining and their boiling ranges. Illustrate: Cracking - hydro cracking, thermal cracking, catalytic cracking, fixed bed and fluidized bed - Catalytic reforming Illustrate: Visbreaking operation - Coking - hydro treating Illustrate: Sulphuric acid Alkylation process, Low temperature Isomerisation



process and Catalytic polymerization processes. using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

## Engineering or professional relevance

In Polytechnic practice, 4 understanding polymerization processes block diagram of a modern refinery operation (including adu, vdu, gas processing, hydrotreater, fccu, isomeriser, alkylation, blending, dcu and sulphur recovery unit) illustrate crude oil fractionation by distillation- crude preheater- atmospheric distillation unit- vacuum distillation unit-petroleum products - list the petroleum products from crude oil refining and their boiling ranges. illustrate: cracking - hydro cracking, thermal cracking, catalytic cracking, fixed bed and fluidized bed - catalytic reforming illustrate: visbreaking operation - coking - hydro treating illustrate: sulphuric acid alkylation process, low temperature isomerisation process and catalytic polymerization processes. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What 4 Understanding polymerization processes Block diagram of a modern refinery operation (Including ADU, VDU, gas processing, hydrotreater, FCCU, isomeriser, alkylation, blending, DCU and sulphur recovery unit) Illustrate Crude oil fractionation by distillation- crude preheater- atmospheric distillation unit- vacuum distillation unit-petroleum products - List the petroleum products from crude oil refining and their boiling ranges. Illustrate: Cracking - hydro cracking, thermal cracking, catalytic cracking, fixed bed and fluidized bed - Catalytic reforming Illustrate: Visbreaking operation - Coking - hydro treating Illustrate: Sulphuric acid Alkylation process, Low temperature Isomerisation process and Catalytic polymerization processes. means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2- 4 Understanding polymerization processes  
Block diagram of a modern refinery operation (Including ADU, VDU, gas processing,

hydrotreater, FCCU, isomeriser, alkylation, blending, DCU and sulphur recovery unit) Illustrate Crude oil fractionation by distillation- crude preheater- atmospheric distillation unit- vacuum distillation unit-petroleum products - List the petroleum products from crude oil refining and their boiling ranges. Illustrate: Cracking - hydro cracking, thermal cracking, catalytic cracking, fixed bed and fluidized bed - Catalytic reforming Illustrate: Visbreaking operation - Coking - hydro treating Illustrate: Sulphuric acid Alkylation process, Low temperature Isomerisation process and Catalytic polymerization processes. **Definition/meaning** **Steps, application** **Technical terms English-**

### Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

Illustrate the downstream processes for crude petroleum.

|               |                                                  |   |
|---------------|--------------------------------------------------|---|
| M2.01         | Illustrate modern refinery operations.           | 1 |
| Understanding | Illustrate petroleum products from crude oil by  |   |
| M2.02         |                                                  | 4 |
| Understanding | fractional distillation.                         |   |
|               | Outline various cracking and reforming processes |   |
| M2.03         |                                                  | 4 |
| Understanding | of petroleum fractions                           |   |
|               | Summarise Visbreaking, Coking, hydro treating    |   |
| M2.04         |                                                  | 4 |
| Understanding | processes.                                       |   |
|               | Summarise Alkylation, Isomerisation and          |   |
| M2.05         |                                                  | 4 |
| Understanding | polymerization processes                         |   |
|               | Series Test – I                                  |   |

Contents:  
 Block diagram of a modern refinery operation (Including ADU, VDU, gas processing, hydrotreater, FCCU, isomeriser, alkylation, blending, DCU and sulphur recovery unit)  
 Illustrate Crude oil fractionation by distillation- crude preheater- atmospheric distillation

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

### MODULE 3

## Summarize the treatment techniques for petroleum fractions.

This module is presented from the official syllabus scope for Petroleum Refining &. Use the topic cards for learning and the source transcript for exact coverage.

**As specified**  
official hours

**CO-linked**  
see outcomes

### Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

### Key facts

- Official module title: Summarize the treatment techniques for petroleum fractions.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

## 2 Understanding petroleum fractions. Summarize the purification and treatment

2 Understanding petroleum fractions. Summarize the purification and treatment is an official topic in Module 3 of Petroleum Refining &. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 2 Understanding petroleum fractions. Summarize the purification and treatment using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 2 understanding petroleum fractions. summarize the purification and treatment should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                                                         |
|---------------------|-------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What 2 Understanding petroleum fractions. Summarize the purification and treatment means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                 |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                       |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3-2 Understanding petroleum fractions.

Summarize the purification and treatment ഹൈഡ്രോക്രൂയിംഗ് definition/meaning ഹൈഡ്രോക്രൂയിംഗ്. ഹൈഡ്രോക്രൂയിംഗ് ഹൈഡ്രോക്രൂയിംഗ്, steps, application ഹൈഡ്രോക്രൂയിംഗ് ഹൈഡ്രോക്രൂയിംഗ്. Technical terms English-ഹൈഡ്രോക്രൂയിംഗ് ഹൈഡ്രോക്രൂയിംഗ്.

## 6 Understanding processes of petroleum fractions. Summarize the important properties of petroleum

6 Understanding processes of petroleum fractions. Summarize the important properties of petroleum is an official topic in Module 3 of Petroleum Refining &. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 6 Understanding processes of petroleum fractions. Summarize the important properties of petroleum using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 6 understanding processes of petroleum fractions. summarize the important properties of petroleum should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                                                                             |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What 6 Understanding processes of petroleum fractions. Summarize the important properties of petroleum means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                                     |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                                           |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3-6 Understanding processes of petroleum fractions. Summarize the important properties of petroleum ഏകദേശം ൧൦൦൦൦ definition/meaning എഴുതുക. ഉപയോഗം ഉൾപ്പെടെ, steps, application ഉൾപ്പെടെ എഴുതുക. Technical terms English-എഴുതുക.

## 2 Understanding products. Outline the experimental methods for determining

2 Understanding products. Outline the experimental methods for determining is an official topic in Module 3 of Petroleum Refining &. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 2 Understanding products. Outline the experimental methods for determining using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 2 understanding products. outline the experimental methods for determining should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                                                      |
|---------------------|----------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What 2 Understanding products. Outline the experimental methods for determining means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                              |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                    |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related



but different term.

**Malayalam support:** Module 3- 2 Understanding products. Outline the experimental methods for determining പദപരിവർത്തനം പദപരിവർത്തനം definition/meaning പദപരിവർത്തനം. പദപരിവർത്തനം പദപരിവർത്തനം, steps, application പദപരിവർത്തനം പദപരിവർത്തനം പദപരിവർത്തനം. Technical terms English- പദപരിവർത്തനം പദപരിവർത്തനം.

**5 Understanding the properties of petroleum products. Physical and chemical impurities in petroleum fractions. Purification and treatment process - sweetening process of petroleum fractions - LPG Merox treatment - hydrodesulphurization of gasoline, kerosene, and diesel - treatment for lube oil. Properties of petroleum fractions (API gravity - Watson Characterization factor - Viscosity - Sulfur content - True boiling point (TBP) curve - Pour point - Flash and fire point - ASTM distillation curve - Knocking -Octane number, Cetane number - Smoke point) Experimental methods for the measurement of viscosity, flash and fire point, cloud point, pour point. Summarize the manufacturing processes and applications of various**

5 Understanding the properties of petroleum products. Physical and chemical impurities in petroleum fractions. Purification and treatment process - sweetening process of petroleum fractions - LPG Merox treatment - hydrodesulphurization of gasoline, kerosene, and diesel – treatment for lube oil. Properties of petroleum fractions (API gravity - Watson Characterization factor - Viscosity - Sulfur content - True boiling point (TBP) curve - Pour point - Flash and fire point - ASTM distillation curve - Knocking -Octane number, Cetane number - Smoke point) Experimental methods for the measurement of viscosity, flash and fire point, cloud point, pour point. Summarize the manufacturing processes and applications of various is an official topic in Module 3 of Petroleum Refining &. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### **Concept and explanation**

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### **Study points**

- Define or identify 5 Understanding the properties of petroleum products. Physical and chemical impurities in petroleum fractions. Purification and treatment process - sweetening process of petroleum fractions - LPG Merox treatment - hydrodesulphurization of gasoline, kerosene, and diesel – treatment for lube oil. Properties of petroleum fractions (API gravity - Watson Characterization factor - Viscosity - Sulfur content - True boiling point (TBP) curve - Pour point - Flash and fire point - ASTM distillation curve - Knocking -Octane number, Cetane number - Smoke point) Experimental methods for the measurement of viscosity, flash and fire point, cloud point, pour point. Summarize the manufacturing processes and applications of various using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

## Engineering or professional relevance

In Polytechnic practice, 5 understanding the properties of petroleum products. physical and chemical impurities in petroleum fractions. purification and treatment process - sweetening process of petroleum fractions - lpg mercox treatment - hydrodesulphurization of gasoline, kerosene, and diesel - treatment for lube oil. properties of petroleum fractions (api gravity - watson characterization factor - viscosity - sulfur content - true boiling point (tbp) curve - pour point - flash and fire point - astm distillation curve - knocking -octane number, cetane number - smoke point) experimental methods for the measurement of viscosity, flash and fire point, cloud point, pour point. summarize the manufacturing processes and applications of various should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What 5 Understanding the properties of petroleum products. Physical and chemical impurities in petroleum fractions. Purification and treatment process - sweetening process of petroleum fractions - LPG Merox treatment - hydrodesulphurization of gasoline, kerosene, and diesel - treatment for lube oil. Properties of petroleum fractions (API gravity - Watson Characterization factor - Viscosity - Sulfur content - True boiling point (TBP) curve - Pour point - Flash and fire point - ASTM distillation curve - Knocking -Octane number, Cetane number - Smoke point) Experimental methods for the measurement of viscosity, flash and fire point, cloud point, pour point. Summarize the manufacturing processes and applications of various means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3- 5 Understanding the properties of petroleum products. Physical and chemical impurities in petroleum fractions. Purification and treatment process - sweetening process of petroleum fractions - LPG Merox treatment - hydrodesulphurization of gasoline, kerosene, and diesel - treatment for

lube oil. Properties of petroleum fractions (API gravity - Watson Characterization factor - Viscosity - Sulfur content - True boiling point (TBP) curve - Pour point - Flash and fire point - ASTM distillation curve - Knocking - Octane number, Cetane number - Smoke point) Experimental methods for the measurement of viscosity, flash and fire point, cloud point, pour point. Summarize the manufacturing processes and applications of various **petroleum fractions** definition/meaning **petroleum fractions**. **petroleum fractions** **petroleum fractions**, steps, application **petroleum fractions** **petroleum fractions**. Technical terms English-**petroleum fractions** **petroleum fractions**.

## Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

Summarize the treatment techniques for petroleum fractions.  
Identify the physical and chemical impurities in

M3.01 Understanding 2

petroleum fractions.

Summarize the purification and treatment

M3.02 Understanding 6

processes of petroleum fractions.

Summarize the important properties of petroleum

M3.03 Understanding 2

products.

Outline the experimental methods for determining

M3.04 Understanding 5

the properties of petroleum products.

Contents:

Physical and chemical impurities in petroleum fractions.

Purification and treatment process - sweetening process of petroleum fractions - LPG Merox

treatment - hydrodesulphurization of gasoline, kerosene, and diesel – treatment for lube oil.

Properties of petroleum fractions (API gravity - Watson Characterization factor - Viscosity -

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam

## MODULE 4

# petrochemicals.

This module is presented from the official syllabus scope for Petroleum Refining &. Use the topic cards for learning and the source transcript for exact coverage.

**As specified**  
official hours

**CO-linked**  
see outcomes

### Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

### Key facts

- Official module title: petrochemicals.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

## Outline the feedstock for petrochemicals.

Outline the feedstock for petrochemicals. is an official topic in Module 4 of Petroleum Refining &. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Outline the feedstock for petrochemicals. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, outline the feedstock for petrochemicals. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                     |
|---------------------|-------------------------------------------------------------------------------------|
| Definition/identity | What Outline the feedstock for petrochemicals. means in the official course context |
| Study lens          | Principle → components or stages → result → application                             |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate   |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4- Outline the feedstock for petrochemicals.

日本語の定義/意味を説明してください。 日本語の 手順、 適用、 技術用語の英語-日本語の 説明をください。

## Summarize the applications of petrochemicals 3 Understanding Illustrate the chemicals produced from methane,

Summarize the applications of petrochemicals 3 Understanding Illustrate the chemicals produced from methane, is an official topic in Module 4 of Petroleum Refining &. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Summarize the applications of petrochemicals 3 Understanding Illustrate the chemicals produced from methane, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, summarize the applications of petrochemicals 3 understanding illustrate the chemicals produced from methane, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                                                                                        |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What Summarize the applications of petrochemicals 3 Understanding Illustrate the chemicals produced from methane, means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                                                |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                                                      |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.



**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-  
Summarize the applications of petrochemicals 3  
Understanding Illustrate the chemicals produced from methane,   
definition/meaning . . . , steps, application . . . . Technical terms English-  
 .

### 3 Understanding synthesis gas and olefins.

3 Understanding synthesis gas and olefins. is an official topic in Module 4 of Petroleum Refining &. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

#### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

#### Study points

- Define or identify 3 Understanding synthesis gas and olefins. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

#### Engineering or professional relevance

In Polytechnic practice, 3 understanding synthesis gas and olefins. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                      |
|---------------------|--------------------------------------------------------------------------------------|
| Definition/identity | What 3 Understanding synthesis gas and olefins. means in the official course context |
| Study lens          | Principle → components or stages → result → application                              |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate    |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-3 Understanding synthesis gas and olefins.

എന്നതിന്റെ definition/meaning എഴുതുക. എന്നതിന്റെ steps, application എന്നിവ എഴുതുക. Technical terms English-ൽ എഴുതുക.

**Comprehend the chemicals from aromatics. 4 Understanding Feedstock for petrochemicals - List the major Gas, Liquid and Solid feedstock's for petrochemical industry - Identify the products from above feedstock's. List the major applications of petrochemicals - polyethylene, polypropylene, polyester, polybutadiene, polyacrylate. Production of methanol by oxidation of methane - Methanol from synthesis gas - production of formaldehyde from methanol. Illustrate the manufacturing of ethylene and propylene by thermal cracking - manufacturing of polyethylene - List the propylene derivatives - Illustrate the production of Acrylic Acid and Methyl Methacrylate (MMA). Illustrate the Manufacturing process for Phenol, Polystyrene Group activity ( Prepare a report on the manufacturing of Caprolactam in FACT and compare the production method with other available manufacturing methods) Text / Reference T/R Book Title/Author T1 Bulk Chemicals from Petroleum; B.K. Bhaskarao R1 Modern Petroleum Refining Processes; B. K. Bhaskarao R2 Petroleum Refinery Engineering; W.L. Nelson, R3 Elements of Petroleum Refinery Engineering; O.P. Gupta R4 Elements of Petrochemical Engineering; Saikat Maitra & O.P. Gupta Online Resources Sl.No Website Link 1 <https://nptel.ac.in/courses/103/102/103102022/>**

Comprehend the chemicals from aromatics. 4 Understanding Feedstock for petrochemicals - List the major Gas, Liquid and Solid feedstock's for petrochemical industry - Identify the products from above feedstock's. List the major applications of petrochemicals - polyethylene, polypropylene, polyester, polybutadiene, polyacrylate. Production of methanol by oxidation of methane - Methanol from synthesis gas - production of formaldehyde from methanol. Illustrate the manufacturing of ethylene and propylene by thermal cracking - manufacturing of polyethylene - List the propylene derivatives - Illustrate the production of Acrylic Acid and Methyl Methacrylate (MMA). Illustrate the Manufacturing process for Phenol, Polystyrene Group activity ( Prepare a report on the manufacturing of Caprolactam in FACT and compare the production method with other available manufacturing methods) Text / Reference T/R Book Title/Author T1 Bulk Chemicals from Petroleum; B.K. Bhaskarao R1 Modern Petroleum Refining Processes; B. K. Bhaskarao R2 Petroleum Refinery Engineering; W.L. Nelson, R3 Elements of Petroleum Refinery Engineering; O.P. Gupta R4 Elements of Petrochemical Engineering; Saikat Maitra & O.P. Gupta Online Resources Sl.No Website Link 1 <https://nptel.ac.in/courses/103/102/103102022/> is an official topic in Module 4 of Petroleum Refining &. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

**Concept and explanation**

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Comprehend the chemicals from aromatics. 4 Understanding Feedstock for petrochemicals – List the major Gas, Liquid and Solid feedstock's for petrochemical industry - Identify the products from above feedstock's. List the major applications of petrochemicals - polyethylene, polypropylene, polyester, polybutadiene, polyacrylate. Production of methanol by oxidation of methane – Methanol from synthesis gas – production of formaldehyde from methanol. Illustrate the manufacturing of ethylene and propylene by thermal cracking – manufacturing of polyethylene - List the propylene derivatives – Illustrate the production of Acrylic Acid and Methyl Methacrylate (MMA). Illustrate the Manufacturing process for Phenol, Polystyrene Group activity ( Prepare a report on the manufacturing of Caprolactam in FACT and compare the production method with other available manufacturing methods) Text / Reference T/R Book  
Title/Author T1 Bulk Chemicals from Petroleum; B.K. Bhaskarao R1 Modern Petroleum Refining Processes; B. K. Bhaskarao R2 Petroleum Refinery Engineering; W.L. Nelson, R3 Elements of Petroleum Refinery Engineering; O.P. Gupta R4 Elements of Petrochemical Engineering; Saikat Maitra & O.P. Gupta Online Resources Sl.No Website Link 1 <https://nptel.ac.in/courses/103/102/103102022/> using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, comprehend the chemicals from aromatics. 4 understanding feedstock for petrochemicals – list the major gas, liquid and solid feedstock's for petrochemical industry - identify the products from above feedstock's. list the major applications of petrochemicals - polyethylene, polypropylene, polyester, polybutadiene, polyacrylate. production of methanol by oxidation of methane – methanol from synthesis gas – production of formaldehyde from methanol. illustrate the manufacturing of ethylene and propylene by thermal cracking – manufacturing of polyethylene - list the propylene derivatives – illustrate the production of acrylic acid and methyl methacrylate (mma). illustrate the manufacturing process for phenol, polystyrene group activity ( prepare a report on the manufacturing of caprolactam in fact and compare the production method with other available manufacturing methods) text / reference t/r book title/author t1 bulk chemicals from petroleum; b.k. bhaskarao r1 modern

petroleum refining processes; b. k. bhaskarao r2 petroleum refinery engineering; w.l. nelson, r3 elements of petroleum refinery engineering; o.p. gupta r4 elements of petrochemical engineering; saikat maitra & o.p. gupta online resources sl.no website link 1 <https://nptel.ac.in/courses/103/102/103102022/> should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

#### Topic study guide

| Aspect              | What to prepare                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What Comprehend the chemicals from aromatics. 4 Understanding Feedstock for petrochemicals - List the major Gas, Liquid and Solid feedstock's for petrochemical industry - Identify the products from above feedstock's. List the major applications of petrochemicals - polyethylene, polypropylene, polyester, polybutadiene, polyacrylate. Production of methanol by oxidation of methane - Methanol from synthesis gas - production of formaldehyde from methanol. Illustrate the manufacturing of ethylene and propylene by thermal cracking - manufacturing of polyethylene - List the propylene derivatives - Illustrate the production of Acrylic Acid and Methyl Methacrylate (MMA). Illustrate the Manufacturing process for Phenol, Polystyrene Group activity ( Prepare a report on the manufacturing of Caprolactam in FACT and compare the production method with other available manufacturing methods) Text / Reference T/R Book Title/Author T1 Bulk Chemicals from Petroleum; B.K. Bhaskarao R1 Modern Petroleum Refining Processes; B. K. Bhaskarao R2 Petroleum Refinery Engineering; W.L. Nelson, R3 Elements of Petroleum Refinery Engineering; O.P. Gupta R4 Elements of Petrochemical Engineering; Saikat Maitra & O.P. Gupta Online Resources Sl.No Website Link 1 <a href="https://nptel.ac.in/courses/103/102/103102022/">https://nptel.ac.in/courses/103/102/103102022/</a> means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-□□ Comprehend the chemicals from aromatics. 4 Understanding Feedstock for petrochemicals - List the major Gas, Liquid and Solid feedstock's for petrochemical industry - Identify the products from above feedstock's. List the major applications of petrochemicals - polyethylene, polypropylene, polyester, polybutadiene, polyacrylate. Production of methanol by oxidation of methane - Methanol from synthesis gas - production of formaldehyde from methanol. Illustrate the manufacturing of ethylene and propylene by thermal cracking - manufacturing of polyethylene - List the propylene derivatives - Illustrate the production of Acrylic Acid and Methyl Methacrylate (MMA). Illustrate the Manufacturing process for Phenol, Polystyrene Group activity ( Prepare a report on the manufacturing of Caprolactam in FACT and compare the production method with other available manufacturing methods) Text / Reference T/R Book Title/Author

T1 Bulk Chemicals from Petroleum; B.K. Bhaskarao R1 Modern Petroleum Refining Processes; B. K. Bhaskarao R2 Petroleum Refinery Engineering; W.L. Nelson, R3 Elements of Petroleum Refinery Engineering; O.P. Gupta R4 Elements of Petrochemical Engineering; Saikat Maitra & O.P. Gupta Online Resources Sl.No Website Link 1 <https://nptel.ac.in/courses/103/102/103102022/>   
 definition/meaning . . . , steps, application . . . . Technical terms English- . . . .

## Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

petrochemicals.

M4.01 Outline the feedstock for petrochemicals. 4

Understanding

M4.02 Summarize the applications of petrochemicals 3

Understanding

Illustrate the chemicals produced from methane,

M4.03 3

Understanding

synthesis gas and olefins.

M4.04 Comprehend the chemicals from aromatics. 4

Understanding

### Series Test – II

Contents:

Feedstock for petrochemicals – List the major Gas, Liquid and Solid feedstock's for

petrochemical industry - Identify the products from above feedstock's.

List the major applications of petrochemicals - polyethylene, polypropylene, polyester, polybutadiene, polyacrylate.

Production of methanol by oxidation of methane – Methanol from synthesis gas – production of formaldehyde from methanol. Illustrate the manufacturing of ethylene and

propylene by thermal cracking – manufacturing of polyethylene - List the propylene

derivatives – Illustrate the production of Acrylic Acid and Methyl Methacrylate

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam

## RAPID REFERENCE

### Concept and formula bank

#### Answer structure

Definition → Principle → Components/steps → Application → Conclusion

Use this sequence for descriptive questions when the topic does not require a numerical formula.

#### Practical record

Aim → Apparatus → Procedure → Observation → Result → Safety

Use this sequence for laboratory, workshop and field activities.

## RAPID REVISION

### Diagram and source-transcript library

#### Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

#### Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

#### Module 3 source and topic cards



Jump to official terminology, topic explanations and study guidance.

## Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

### OFFICIAL SELF-LEARNING

## Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

### EXAMINATION PREPARATION

## Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

The official assessment structure is reproduced from the supplied PDF.

**Practice-paper notice:** The model paper below is generated for revision and is not an official examination paper.

### ACTIVE RECALL

## Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

### Module 1

**Define or explain Classify fossil fuels. 2 Remembering Summarize the production methods,. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Classify fossil fuels. 2 Remembering Summarize the production methods,*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

**Define or explain 4 Understanding characteristics and properties of gaseous fuels. Summarise the characterization and properties of. (3 marks)**

**Answer:** Begin with the standard definition or identity of *4 Understanding characteristics and properties of gaseous fuels. Summarise the characterization and properties of.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

**Define or explain 4 Understanding crude petroleum. Illustrate the upstream processes for crude. (3 marks)**

**Answer:** Begin with the standard definition or identity of *4 Understanding crude petroleum. Illustrate the upstream processes for crude.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

**Define or explain 4 Understanding petroleum. Classification of fuel- Categorize primary & secondary solid, liquid and gaseous fuels with examples. Identify the composition, properties and uses of Natural gas, - Illustrate the**

**production of natural gas - Comprehend the production of shale gas. Origin& formation of petroleum - Classification of crude oil - Properties of crude oil - API gravity and density, viscosity, sulphur content and vapour pressure - Evaluation of crude oil. Exploration and drilling of crude oil - Gas-oil separation - Dehydration - Desalting- Crude stabilization and Enhanced oil recovery - Sources of indigenous crude - Sources of imported crude - Petroleum industries in India. (3 marks)**

**Answer:** Begin with the standard definition or identity of *4 Understanding petroleum. Classification of fuel- Categorize primary & secondary solid, liquid and gaseous fuels with examples. Identify the composition, properties and uses of Natural gas, - Illustrate the production of natural gas - Comprehend the production of shale gas. Origin& formation of petroleum - Classification of crude oil - Properties of crude oil -API gravity and density, viscosity, sulphur content and vapour pressure - Evaluation of crude oil. Exploration and drilling of crude oil - Gas-oil separation - Dehydration - Desalting- Crude stabilization and Enhanced oil recovery - Sources of indigenous crude - Sources of imported crude - Petroleum industries in India.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

## Module 2

**Define or explain Illustrate modern refinery operations. 1 Understanding Illustrate petroleum products from crude oil by. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Illustrate modern refinery operations. 1 Understanding Illustrate petroleum products from crude oil by.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

**Define or explain 4 Understanding fractional distillation. Outline various cracking and reforming processes. (3 marks)**

**Answer:** Begin with the standard definition or identity of 4 *Understanding fractional distillation. Outline various cracking and reforming processes.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

**Define or explain 4 Understanding of petroleum fractions Summarise Visbreaking, Coking, hydro treating. (3 marks)**

**Answer:** Begin with the standard definition or identity of 4 *Understanding of petroleum fractions Summarise Visbreaking, Coking, hydro treating.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

**Define or explain 4 Understanding processes. Summarise Alkylation, Isomerisation and. (3 marks)**

**Answer:** Begin with the standard definition or identity of 4 *Understanding processes. Summarise Alkylation, Isomerisation and.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

## Module 3

**Define or explain 2 Understanding petroleum fractions. Summarize the purification and treatment. (3 marks)**

**Answer:** Begin with the standard definition or identity of 2 *Understanding petroleum fractions. Summarize the purification and treatment.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

**Define or explain 6 Understanding processes of petroleum fractions. Summarize the important properties of petroleum. (3 marks)**

**Answer:** Begin with the standard definition or identity of 6 *Understanding processes of petroleum fractions. Summarize the important properties of petroleum.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

**Define or explain 2 Understanding products. Outline the experimental methods for determining. (3 marks)**

**Answer:** Begin with the standard definition or identity of 2 *Understanding products. Outline the experimental methods for determining.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

**Define or explain 5 Understanding the properties of petroleum products. Physical and chemical impurities in petroleum fractions. Purification and treatment process - sweetening process of petroleum fractions - LPG Merox treatment - hydrodesulphurization of gasoline, kerosene, and diesel - treatment for lube oil. Properties of petroleum fractions (API gravity - Watson Characterization factor - Viscosity - Sulfur content - True boiling point (TBP)**

**curve - Pour point - Flash and fire point - ASTM distillation curve - Knocking - Octane number, Cetane number - Smoke point) Experimental methods for the measurement of viscosity, flash and fire point, cloud point, pour point. Summarize the manufacturing processes and applications of various. (3 marks)**

**Answer:** Begin with the standard definition or identity of *5 Understanding the properties of petroleum products. Physical and chemical impurities in petroleum fractions. Purification and treatment process - sweetening process of petroleum fractions - LPG Merox treatment - hydrodesulphurization of gasoline, kerosene, and diesel - treatment for lube oil. Properties of petroleum fractions (API gravity - Watson Characterization factor - Viscosity - Sulfur content - True boiling point (TBP) curve - Pour point - Flash and fire point - ASTM distillation curve - Knocking - Octane number, Cetane number - Smoke point) Experimental methods for the measurement of viscosity, flash and fire point, cloud point, pour point. Summarize the manufacturing processes and applications of various. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.*

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

## Module 4

**Define or explain Outline the feedstock for petrochemicals.. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Outline the feedstock for petrochemicals..* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

**Define or explain Summarize the applications of petrochemicals 3 Understanding Illustrate the chemicals produced from methane,. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Summarize the applications of petrochemicals 3 Understanding Illustrate the chemicals produced from methane,.* State

the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

### Define or explain 3 Understanding synthesis gas and olefins.. (3 marks)

**Answer:** Begin with the standard definition or identity of *3 Understanding synthesis gas and olefins..* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

**Define or explain Comprehend the chemicals from aromatics. 4 Understanding Feedstock for petrochemicals - List the major Gas, Liquid and Solid feedstock's for petrochemical industry - Identify the products from above feedstock's. List the major applications of petrochemicals - polyethylene, polypropylene, polyester, polybutadiene, polyacrylate. Production of methanol by oxidation of methane - Methanol from synthesis gas - production of formaldehyde from methanol. Illustrate the manufacturing of ethylene and propylene by thermal cracking - manufacturing of polyethylene - List the propylene derivatives - Illustrate the production of Acrylic Acid and Methyl Methacrylate (MMA). Illustrate the Manufacturing process for Phenol, Polystyrene Group activity ( Prepare a report on the manufacturing of Caprolactam in FACT and compare the production method with other available manufacturing methods) Text / Reference T/R Book Title/Author T1 Bulk Chemicals from Petroleum; B.K. Bhaskarao R1 Modern Petroleum Refining Processes; B. K. Bhaskarao R2 Petroleum Refinery Engineering; W.L. Nelson, R3 Elements of Petroleum Refinery Engineering; O.P. Gupta R4 Elements of Petrochemical Engineering; Saikat Maitra & O.P. Gupta Online Resources Sl.No Website Link 1 <https://nptel.ac.in/courses/103/102/103102022/>. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Comprehend the chemicals from aromatics. 4 Understanding Feedstock for petrochemicals - List the major Gas, Liquid and Solid feedstock's for petrochemical industry - Identify the products from*

above feedstock's. List the major applications of petrochemicals - polyethylene, polypropylene, polyester, polybutadiene, polyacrylate. Production of methanol by oxidation of methane - Methanol from synthesis gas - production of formaldehyde from methanol. Illustrate the manufacturing of ethylene and propylene by thermal cracking - manufacturing of polyethylene - List the propylene derivatives - Illustrate the production of Acrylic Acid and Methyl Methacrylate (MMA). Illustrate the Manufacturing process for Phenol, Polystyrene Group activity ( Prepare a report on the manufacturing of Caprolactam in FACT and compare the production method with other available manufacturing methods) Text / Reference T/R Book Title/Author T1 Bulk Chemicals from Petroleum; B.K. Bhaskarao R1 Modern Petroleum Refining Processes; B. K. Bhaskarao R2 Petroleum Refinery Engineering; W.L. Nelson, R3 Elements of Petroleum Refinery Engineering; O.P. Gupta R4 Elements of Petrochemical Engineering; Saikat Maitra & O.P. Gupta Online Resources Sl.No Website Link 1 <https://nptel.ac.in/courses/103/102/103102022/>. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

**Malayalam support:** □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

## PRACTICE ONLY

# Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

## Module 1

1-mark definition: State the meaning of **Classify fossil fuels. 2 Remembering Summarize the production methods,**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.



## Module 2

1-mark definition: State the meaning of **Illustrate modern refinery operations. 1 Understanding Illustrate petroleum products from crude oil by.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

## Module 3

1-mark definition: State the meaning of **2 Understanding petroleum fractions. Summarize the purification and treatment.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

## Module 4

1-mark definition: State the meaning of **Outline the feedstock for petrochemicals..**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

### QUICK REVISION

## Exam checklist

### Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.

- Plan labels, units, sequence and marks before writing.

### Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

## REFERENCE VOCABULARY

### Glossary

#### Study glossary

| Term              | Meaning in this lesson                                                   |
|-------------------|--------------------------------------------------------------------------|
| Course Outcome    | A capability the student should demonstrate after completing the course. |
| Module            | An official syllabus unit with defined topics and hours.                 |
| CIE               | Continuous Internal Evaluation.                                          |
| ESE               | End Semester Examination.                                                |
| Source transcript | A preserved extract of the official syllabus used for coverage checking. |

## SOURCE-SUPPORTED REFERENCES

### References and web resources

#### References listed in the official PDF

References are included in the official source PDF.

#### Web resources listed in the official PDF

- Sl.No Website Link
- <https://nptel.ac.in/courses/103/102/103102022/>

## IN-PAGE SEARCH

## Search this lesson

Prepared from the supplied official SITTTR syllabus PDF for Course 5073A. Verify institutional updates against the current official curriculum.